

Examples of Asset/Liability Management

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Although it has evolved to reflect changing circumstances in the economy and markets, in its simplest form, asset/liability management involves managing assets and cash flows to satisfy obligations. It is a form of risk management in which the investor seeks to mitigate or hedge the risk of failing to meet liability obligations. Success should increase the profitability of the organization, in addition to managing risk.

Some practitioners prefer the phrase "surplus optimization" to explain the need to maximize assets to meet increasingly complex liabilities. Alternatively, the surplus is also known as net worth or the difference between the market value of assets and the present value of liabilities. Asset and liability management is conducted from a long-term perspective that manages risks arising from the accounting of assets vs. liabilities. As such, it can be both strategic and tactical.

A monthly mortgage is a common example of a liability that a consumer pays for from current cash inflows. Each month, the mortgagor must have sufficient assets to pay their mortgage. Institutions have similar challenges but on a much more complex scale. For example, a pension plan must contractually satisfy established benefit payments to retirees while sustaining an asset base through prudent asset allocation and risk monitoring from which to generate future ongoing payments.

The liabilities of institutions are complex and varied. The challenge is to understand their characteristics and structure assets in a strategic and complementary way. This may result in an asset allocation that would appear sub-optimal (if only assets were being considered). Assets and liabilities are usually thought of as intricately intertwined rather than separate concepts. Here are some examples of the asset/liability challenges of institutions and individuals.

Key Takeaways

- The need for asset/liability management can arise in a variety of situations, scenarios, and industries.
- Asset/liability management can also be referred to as liability driven investing.
- In any scenario, asset/liability management involves ensuring that assets are available to appropriately cover liabilities when they are due or expected to be due.

The Banking Industry

As a financial intermediary, banks accept deposits for which they are obligated to pay interest (liabilities) and offer loans for which they receive interest (assets). In addition to loans, security portfolios also compose bank assets. Banks must manage interest rate

risk, which can lead to a mismatch of assets and liabilities. Volatile interest rates and the abolition of Regulation Q, which capped the rate at which banks could pay depositors, contributed to this problem.

A bank's net interest margin—the difference between the rate that it pays on deposits and the rate that it receives on its assets (loans and securities)—is a function of interest rate sensitivity and the volume and mix of assets and liabilities. To the extent that a bank borrows in the short term and lends for the long term, there is often a mismatch that the bank must address through the structuring of its assets and liabilities or with the use of derivatives (e.g., swaps, swaptions, options, and futures) to ensure it satisfies all of its liabilities.

Insurance Companies

There are two main types of insurance companies: life and non-life (e.g., property and casualty). Life insurers also offer annuities that may be life or non-life contingent, guaranteed rate accounts (GICs), or stable value funds.

Life insurance tends to be a longer-term liability. A life insurance policy varies by type but the standard is usually based around paying out a lump sum to a beneficiary after the death of an owner. This requires actuarial planning using life expectancy tables and other factors to determine the estimated annual obligations that an insurer will likely face each year.

With annuities, liability requirements entail funding income for the duration of a payout period that begins on a specified date. For GICs and stable value products, they are subject to interest rate risk, which can erode a surplus and cause assets and liabilities to be mismatched. Liabilities of life insurers tend to be longer in duration. Accordingly, longer duration and inflation-protected assets are selected to match those of the liability (longer maturity bonds and real estate, equity, and venture capital), although product lines and their requirements vary.

Non-life insurers must meet liabilities (accident claims) of a much shorter duration due to the typical three to five-year underwriting cycle. The business cycle tends to drive a company's need for liquidity. Interest rate risk is less of a consideration for a non-life insurer than a life insurer. Liabilities tend to be uncertain concerning both value and timing. The liability structure of a company is a function of its product line and the claims and settlement process, which often are a function of the so-called "long tail" or period between the occurrence and claim reporting and the actual payout to the policyholder. This arises because commercial clients represent a far larger portion of the total property and casualty market than in the life insurance business, which is mainly a business that caters to individuals.

Insurance companies offer a multitude of products that require extensive plans for asset/liability management by the insurer.

The Benefit Plan

A traditional defined benefit plan must satisfy a promise to pay the benefit formula specified in the plan document of the plan sponsor. Accordingly, investment is long-term in nature, with a view to maintaining or growing the asset base and providing retirement payments. In the practice known as liability-driven investing (LDI), managers gauge the liabilities by estimating the duration of benefit payments and their present value.

Funding a benefit plan often involves matching variable rate assets with variable rate liabilities (future retirement payments based upon salary growth projections of active workers) and fixed-rate assets with fixed-rate liabilities (income payments to retirees). As portfolios and liabilities are sensitive to interest rates, strategies such as portfolio immunization and duration matching may be employed to protect portfolios from rate fluctuations.

Foundations and Non-Profits

Institutions that make grants and are funded by gifts and investments are foundations. Endowments are long-term funds owned by non-profit organizations (e.g., universities and hospitals). They tend to be perpetual in design. Their liability is usually an annual spending commitment as a percentage of the market value of assets. The long-term nature of these arrangements often leads to a more aggressive investment allocation meant to outpace inflation, grow the portfolio, and support and sustain a specific spending policy.

Wealth Management

With private wealth, the nature of individuals' liabilities may be as varied as the individuals themselves. These range from retirement planning and education funding to home purchases and unique circumstances. Taxes and risk preferences will frame the asset allocation and risk management process that determines the appropriate asset allocation to meet these liabilities. Techniques of asset/liability management may be compared to those used on an institutional level, in particular fund strategies used for targeting cash flows after a specified date.

Large Conglomerate, Multi-National Corporations

Finally, corporations can use asset/liability management techniques for all kinds of purposes. Some motivations may include liquidity, foreign exchange, interest rate risks, and commodity risks. An airline for example, might hedge its exposure to fluctuations in fuel prices in order to maintain manageable asset/liability matching. Moreover, multi-national companies might hedge the risks of currency losses through the foreign exchange market in order to ensure they have a better forecast for managing assets vs. payments.

The Bottom Line

Asset/liability management, also known as liability driven investing, can be a complex endeavor. An understanding of the internal and external factors that affect risk management is critical to finding an appropriate solution. Prudent asset allocation accounts not only for the growth of assets but also specifically addresses the nature of an organization's liabilities.